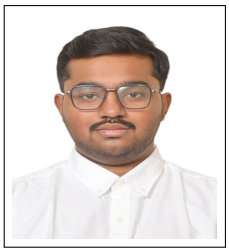
Anant Mukherjee

+91-8347484899
anantmukherjee.che@gmail.com
linkedin.com/in/anant-mukherjee
github.com/voidsyntax-dev



Summary

Chemical engineering undergraduate at NIT Calicut with a focus on process simulation and design. Proficient in Machine Learning and Aspen, with completed projects on Soft Sensor Modeling, Machine Learning driven predictive maintenance, crude distillation and biodiesel production plants. Gained hands-on exposure to refinery operations and chemical manufacturing during internships. Interested in process optimization, sustainable process design, and machine learning applications in plant operations.

Education

National Institute of Technology Calicut — **May 2028**
B.Tech, Chemical Engineering — CGPA: 8.78 / 10 — *Kozhikode, Kerala*

Green Valley High School — **2024**
CBSE Class XII — 93.2% — *Vadodara, Gujarat*

Bright Day School — **2022**
CBSE Class X — 94.8% — *Vadodara, Gujarat*

Experience

Indian Oil Corporation Limited (IOCL) — **May 2026 – Jul. 2026**
Refinery Intern — *Vadodara, Gujarat*

- Analyzed process flow configurations and operating objectives of major refinery units including Crude Distillation (CDU), Vacuum Distillation (VDU), Hydrocracking, Delayed Coking and Fluid Catalytic Cracking (FCC) to understand refinery-wide crude conversion pathways.
- Evaluated crude oil fractionation, product upgrading, and conversion processes while examining the integration of primary and secondary refining operations across the petroleum value chain.
- Mapped material flow pathways and product stream distributions across refinery units, assessing their contribution to the production of transportation fuels and value-added petroleum products.

Jubilant Ingrevia — **Nov. 2025 – Dec. 2025**
Production Intern — *Savli, Gujarat*

- Analyzed the end-to-end production of aqueous choline chloride, covering raw material handling, concentration-based processing, storage, and product routing in a continuous production plant.
- Evaluated process flow and operational considerations for handling ethylene oxide, trimethylamine, and HCl, and assessed nitrogen generation and steam boiler utilities that support plant reliability.

Projects

Soft Sensor for Ethylbenzene Production | *Aspen Dynamics, Python, Time-Series Validation*

- Developed a soft sensor to estimate the outlet composition of an ethylbenzene CSTR, a quantity that is difficult to measure online, from temperatures, pressures, flows, and setpoints. Generated training data from an Aspen Dynamics model using randomized setpoint and feed changes.
- Engineered lag features within continuous data segments and applied a time-ordered 70/30 split to avoid leakage between correlated samples. Compared baseline, Ridge, PLS, random forest, SVR, MLP, and gradient boosting models.
- Selected MLP models reached R^2 of 0.979 for ETBENZ and 0.892 for D-ETBENZ on held-out data, and were saved with their scalers, feature lists, and lag settings for deployment.

Predictive Maintenance of Milling Machine | *Scikit-Learn, Gradient Boosting, Machine Learning*

- Built a gradient boosting failure classifier on 10,000 industrial machine records, using engineered features such as temperature difference, mechanical power, and torque-to-speed ratio.
- Achieved PR-AUC of 0.905 and ROC-AUC of 0.974 on a chronological hold-out set, catching 84% of failures at 95% precision using a leakage-free pipeline.
- Set the decision threshold by expected-cost minimisation, treating a missed failure as ten times costlier than a false alarm, and compared seven resampling strategies to handle the 3.4% failure imbalance.

SkillSetu | *Next.js 15, React 19, Prisma, PostgreSQL, Gemini 1.5 Flash, D3.js*

- Built a full-stack AI career-learning platform with Next.js, React 19, Prisma, PostgreSQL, and D3.js that maps careers to skills, resources, and assessments, with 15+ production-ready modules.
- Designed adaptive algorithms for mastery estimation, Ebbinghaus-based skill decay, and career-readiness scoring, and built a RAG-based AI tutor grounded in curated study resources.

Crude Distillation Unit Simulation | *Aspen HYSYS, Refinery Process Design*

- Simulated a 4.5 MMTPA crude distillation unit in Aspen HYSYS, with a crude preheat train, a fired furnace, a preflash column, two debutanisers to strip light gases from naphtha, and an atmospheric column with three side draws producing kerosene, diesel, and AGO.
- Characterised the crude with a 61-cut assay and closed mass balances across all separation sections to 0.0001%. Quantified 165.6 MW of heat input and identified about 56 MW of heat integration potential from hot atmospheric product streams to the preheat train.

Biodiesel Production Plant Simulation | *Aspen Plus, Chemical Process Design*

- Simulated a continuous alkali-catalysed transesterification plant processing 1,050 kg/h of oil, achieving 96.4% one-pass ester conversion and 94.0% methanol recovery through recycle loops.
- Verified FAME product at 99.7 wt% ester content against EN 14214 limits, with vacuum columns accounting for 75% of duty and net heating of about 125 kW.

Technical Skills

Process Simulation: DWSIM, Aspen HYSYS, Aspen Plus

Programming & Data Analysis: Python, NumPy, Pandas, scikit-learn, Microsoft Excel, MATLAB

Certifications

Introduction to OSHA: Safety Standards and Compliance — *Coursera*

Excel Basics for Data Analysis — *IBM*

Achievements & Recognition

Top 50 Projects Across India — *Atal Innovation Mission (AIM), 2022*

JEE Mains All India Rank 20,972 — *2024*

JEE Advanced All India Rank 20,466 — *2024*

International and Zonal Excellence Awards — *SOF Olympiads, Science and Mathematics*